ITS 2122

Python for Data Science & AI

Group Project Specification

Semester 3, 2026

PROJECT TITLE

Credit Risk & Customer Financial Health Analytics for a Digital Lending Provider

Domain	Finance / Digital Lending / Credit Risk Analytics
Module Component	Final Group Project & Presentation (40% of module grade)
Deliverables	Business Report (PDF) + Technical Appendix (Jupyter Notebook) + Presentation

Part 1: The Business Mandate & Project Overview

1.1 Introduction: Your Role as Data Science Consultants

For this capstone project, your group will operate as a team of data science consultants engaged by the executive board of "**FinTrust Digital Finance**," a fast-growing financial technology lender that issues credit cards and short-term personal credit to salaried customers. The company has acquired a large base of borrowers and accumulated a substantial volume of account and repayment data. However, its credit-monitoring and collections decisions remain largely manual and intuitive rather than data-driven.

The leadership team recognises that to manage risk responsibly, reduce defaults, and protect both the business and its customers, it must treat its data as a strategic asset. Your team's mandate is to analyse historical credit-card account data, uncover the behavioural drivers of repayment and default, and deliver a clear, evidence-based risk-monitoring framework. You are expected to move beyond simple reporting and produce high-value intelligence that can directly inform credit policy, early-warning monitoring, and customer engagement.

1.2 The Core Challenge: From Data to Decisions

The management team has provided you with a portfolio of customer accounts covering demographics, granted credit limits, six months of bill statements, six months of repayments, repayment-status history, and a flag indicating whether each customer defaulted in the following month. They have identified several critical blind spots and have tasked your team with addressing the following fundamental questions:

1. **Default Risk Profiling:** What is the overall default rate, and how does it vary across age bands, education levels, marital status, and credit-limit tiers? Which customer groups carry the greatest risk?
2. **Repayment Behaviour:** How do repayment delays, bill amounts, and payment amounts over the recent months relate to the likelihood of default? Which behaviours are the strongest early-warning signals?
3. **Credit Exposure vs. Capacity:** Where is the company over-exposed — that is, where are high credit limits combined with weak repayment behaviour, and what is the financial significance of these accounts?
4. **Customer Segmentation:** Can we move beyond a one-size-fits-all approach and group customers into meaningful financial-health segments (for example Healthy, Watchlist, At-Risk, Critical) based on their behaviour?
5. **Actionable Strategy:** How should each segment be managed differently through reminders, credit reviews, restructuring offers, or financial-wellness support, while remaining fair and transparent?

1.3 Project Objectives

This project is a comprehensive application of the skills and concepts covered in ITS 2122. Successful completion will require your team to demonstrate proficiency across the full data-analysis lifecycle. Specifically, you will be expected to:

- **Data Ingestion & Cleaning:** Apply robust loading, validation, and preprocessing techniques with Pandas to handle real-world imperfections such as invalid category codes, duplicates, and inconsistent values. This assesses Module 5 (File Handling) and Module 6 (Data Manipulation).
- **Exploratory Data Analysis (EDA):** Conduct a thorough EDA with Pandas and NumPy to dissect repayment behaviour, surface patterns, and answer the board's core questions. This assesses Module 6 (Data Manipulation).

- **Data Visualization:** Produce a portfolio of clear, professionally formatted visualizations with Matplotlib and Seaborn to communicate findings to a non-technical audience. This assesses Module 7 (Data Visualization).
- **Advanced Analytics & Modelling:** Engineer behavioural features and implement an interpretable credit-risk scoring and segmentation framework using Python data structures, user-defined functions, and advanced Pandas operations. This covers Modules 2, 3, and 6.
- **Data Enrichment:** Acquire and integrate external finance data through a public web API using the requests library. This assesses Module 8 (APIs, Data Collection & Debugging).
- **Professional Communication:** Synthesise all findings into a persuasive, professionally structured business report that delivers actionable, evidence-based recommendations.

Part 2: The Primary Asset — Data & Tools

2.1 The Dataset: Default of Credit Card Clients

The foundation of your analysis will be the public "Default of Credit Card Clients" dataset from the UCI Machine Learning Repository [1]. The dataset contains 30,000 customer records from a credit-card issuer, including demographic attributes, granted credit, six months of billing and payment history, and a target variable indicating default in the next month. It is well documented and highly relevant to the business scenario.

Your first task, which directly tests your file-handling capabilities (Module 5), will be to load the provided data file (.csv or .xls) into a Pandas DataFrame to begin your analysis.

2.2 Data Dictionary

A precise understanding of the data is critical for meaningful analysis. The following table summarises the key fields and the nuances your team must account for during cleaning and feature engineering.

Variable	Description	Key Considerations & Notes
ID	Unique identifier for each customer account.	An identifier only — exclude from any statistical or correlation analysis.
LIMIT_BAL	Amount of granted credit (the credit limit), in currency units.	Use for exposure analysis and credit-limit tiers; check for extreme outliers.
SEX, EDUCATION, MARRIAGE	Demographic attributes, stored as numeric codes.	Must be decoded into readable labels. Some codes are undocumented/unknown and must be handled and justified.
AGE	Customer age in years.	Group into age bands for EDA; validate for impossible values.
PAY_1 … PAY_6	Repayment status for the six most recent months.	Core delinquency signal. Positive values indicate months of delay; document the encoding you adopt.
BILL_AMT1 … BILL_AMT6	Bill statement amount for each of the six months.	Used for balance trends and utilisation-style features; values can be zero or negative.
PAY_AMT1 … PAY_AMT6	Amount paid in each of the six months.	Used for repayment consistency and payment-to-bill ratios; guard against divide-by-zero.

Variable	Description	Key Considerations & Notes
default payment next month	Target: 1 if the customer defaulted next month, else 0.	Used to compute default rates and validate the risk segments you build.

Note: In the given dataset, all currency-related data are expressed in **New Taiwan Dollars (NTD/TWD)**.

2.3 Required Python Libraries

To complete this project, you will use the core data-science stack taught in this module. Your Jupyter Notebook environment should import the following essential libraries:

- **pandas:** The primary tool for loading, cleaning, transforming, grouping, and analysing the data.
- **numpy:** For numerical operations and custom feature calculations.
- **matplotlib and seaborn:** The standard libraries for clear, annotated, and aesthetically pleasing visualizations.
- **requests:** Required for the API integration task in Phase 5.

Part 3: The Analytical Blueprint — A Phased Approach

Follow this structured, phased approach to guide your analysis from raw data to strategic recommendations. Each phase builds upon the last and is designed to ensure a comprehensive, methodologically sound project.

3.1 Phase 1 — Data Sanitation and Preprocessing

The quality of your insights depends directly on the quality of your data. This phase transforms the raw dataset into a clean, reliable, analysis-ready format.

1. **Load Data:** Ingest the dataset into a Pandas DataFrame and confirm the row/column counts against the documentation.
2. **Initial Assessment:** Audit the data using .info(), .describe(include='all'), .isnull().sum(), and .value_counts() on categorical fields.
3. **Rename Columns:** Convert columns to clear, Python-friendly names (for example limit_balance, pay_status_1, bill_amount_1, default_next_month).
4. **Decode Categories:** Map the numeric SEX, EDUCATION, and MARRIAGE codes to readable labels, explicitly handling undocumented/unknown codes.
5. **Handle Duplicates & Invalid Values:** Remove duplicate rows and validate ranges for age, repayment status, bills, and payments; state and justify every decision.
6. **Feature Engineering:** Create behavioural features such as average bill amount, average payment amount, payment-to-bill ratio, number of delayed months, maximum delay, and a recent-balance trend indicator.
7. **Persist the Result:** Export the cleaned dataset (for example cleaned_credit_data.csv) to support reproducibility.

3.2 Phase 2 — Exploratory Data Analysis & Insight Generation

With a clean dataset, dive into exploration to uncover patterns and answer the board's questions, moving from observation to interpretation.

1. **Default-Rate Analysis:** Compute the overall default rate and compare it across age bands, education, marital status, and credit-limit tiers using grouped summaries and bar charts.
2. **Repayment-Behaviour Analysis:** Examine repayment-status patterns across the six months and identify which delays most strongly relate to default.
3. **Financial Position Analysis:** Compare bill amounts, payment amounts, and payment-to-bill ratios for defaulted versus non-defaulted customers.
4. **Relationship Analysis:** Use correlation analysis and/or grouped aggregations to investigate how credit limit, bills, payments, and delays relate to default; present a correlation heatmap.
5. **Visual Portfolio:** Produce at least eight meaningful visualizations, each with a clear title, labelled axes, an appropriate chart type, and a short written interpretation.

3.3 Phase 3 — Advanced Analytics: Credit-Risk Segmentation

This phase moves from descriptive to diagnostic analytics. You will build an interpretable credit-risk scoring framework — analogous in rigour to an RFM model — to quantitatively rank and segment customers by financial health.

1. **Define Risk Dimensions:** Select clear behavioural dimensions, for example Delinquency (repayment delay), Repayment Capacity (payment-to-bill ratio), and Exposure (credit limit / outstanding balance).
2. **Score Customers:** For each dimension, use pandas.qcut() or documented rules to assign scores (e.g. 1–5), ensuring the direction is correct (worse behaviour = higher risk).
3. **Combine Scores:** Concatenate or aggregate the dimension scores into an overall risk score implemented through a reusable user-defined function.
4. **Map to Segments:** Use a Python dictionary or function to translate scores into descriptive, business-friendly segments. Validate each segment by checking its actual default rate.

Your segmentation should produce business-interpretable groups similar to the following framework:

Segment	Characteristics
Healthy	Pays on time, strong payment-to-bill ratio, low exposure relative to limit.
Watchlist	Occasional short delays or a weakening payment ratio; early signs of stress.
At-Risk	Repeated delays, high outstanding bills, or low repayment consistency.
Critical / Intervention	Severe and sustained delays with high exposure; highest default rate.

3.4 Phase 4 — Strategic Recommendations

A technical risk model is only useful when its output is translated into action. This phase requires you to think like a credit-risk and customer strategist.

Investigate the Credit Utilisation and Exposure Hypothesis: Some customers may be using a large percentage of their available credit limit and showing weak repayment behaviour. Create a histogram or box plot of credit utilisation, outstanding balance, and credit limit across customer segments. Examine whether customers with high utilisation and poor repayment behaviour contribute disproportionately to default risk. Comment on this finding, because the strategy for a customer using most of their available credit may differ from the strategy for a customer with low utilisation. Translate each segment into specific, fair, and transparent actions such as reminders, credit-limit reviews, payment restructuring, or financial-wellness support.

3.5 Phase 5 — Data Enrichment via API Integration

This task lets your team demonstrate advanced skills in data acquisition and integration, as covered in Module 8.

Select a free, public finance API (for example a currency-exchange API such as ExchangeRate-API, or the World Bank indicators API). Using the requests library, fetch relevant external data — for instance exchange rates to convert credit limits and balances into a reporting currency (USD/EUR), or a macroeconomic indicator such as inflation or lending interest rate for context. Apply the result to your dataset (for example creating LIMIT_BAL_USD) or summarise it alongside your analysis. In a dedicated notebook section, document the API used, show a sample JSON response, handle errors gracefully, and explain the business value of the enrichment (e.g. reporting to international stakeholders or relating risk to macroeconomic conditions). Do not embed any private API keys in the notebook.

Part 4: Final Deliverables

Your submission will consist of **a professional business report**, and **a technical appendix** as summarised below. You should also submit the cleaned dataset file.

Deliverable	Format	Requirement
Strategic Insights Report	PDF	**2,500–3,000 words (strict)**, written for the executive board. Insight-led, not a code dump; visualizations support the narrative. Excludes title page, contents, and appendix from the word count. Your report should begin with an **Executive Summary**. It should consist of a concise, one-page overview of the project's purpose, your most critical findings, and your top three strategic recommendations. This is the most important page of the report. After that, you should describe all the processes and methodologies you followed, the results you obtained, and the conclusions you reached for each phase from **Phase 1 to Phase 5** (i.e., Sections **3.1 to 3.5**).
Technical Appendix	Jupyter Notebook (.ipynb)	A single, well-organised, reproducible notebook following the phased structure, with Markdown explaining the rationale behind each step. Must run top-to-bottom without errors. After each cell, the corresponding output should be clearly visible, including any plots, tables, and other generated results.
Cleaned Dataset	CSV / Excel	The cleaned dataset as a CSV or Excel file..

4.1 The Strategic Insights Report (PDF)

This is the primary deliverable for the executive board. It must be polished and written in clear business language — not a code report. The focus must be on insights and recommendations, with visualizations used to support your narrative. Required structure:

1. Executive Summary — a concise one-page overview of purpose, most critical findings, and top three recommendations.
2. Introduction — the business problem and project objectives as defined in the mandate.
3. Data & Methodology — the dataset and the key steps of your cleaning and feature-engineering approach (Phase 1).
4. Credit Risk & Repayment Analysis — your EDA findings (Phase 2), illustrated with your best visualizations.
5. Customer Segmentation & Strategy — the core analytical section: segment sizes, characteristics, default rates, and the high-exposure findings (Phases 3–4).
6. Data Enrichment via API — the results and business value of your API integration (Phase 5).
7. Recommendations, Ethics & Limitations — actionable strategy, responsible-analytics considerations, and limitations.

4.2 The Technical Appendix (Jupyter Notebook)

Submit a single, well-organised notebook containing all the Python code used for your analysis. It serves as the technical documentation for your project and must be reproducible.

- **Clean & Logical Flow:** Follow the phased structure of the project (Phase 1 to Phase 5).
- **Well-Documented Code:** Code cells must be accompanied by Markdown explaining why each step is performed and how outputs are interpreted.
- **Reproducibility:** The notebook must run from top to bottom without errors, assuming the data file is in the correct location.

Part 5: Assessment & Evaluation Rubric

5.1 Grading Breakdown

The project is graded out of 100 points. Note that marks are not awarded purely on a group-wise basis: the report, notebook, and teamwork are all assessed in light of individual performance in the final presentation and viva, so members of the same group may receive different marks.

Component	Weight
Strategic Insights Report (PDF)	50%
Technical Appendix (Jupyter Notebook)	40%
Teamwork & Collaboration (Peer Evaluation)	10%

5.2 Detailed Rubric

Your submission will be evaluated against the following criteria to ensure a fair and transparent assessment process.

Criteria	Wt.	Unsatisfactory (0–49%)	Satisfactory (50–69%)	Good (70–89%)	Excellent (90–100%)
Code Quality & Correctness	20%	Non-functional, major logical errors, disorganised, or signs of plagiarism.	Runs but is inefficient, poorly structured, and lightly commented.	Functional, mostly correct, with documentation explaining its purpose.	Efficient, well-structured, extensively documented, follows PEP 8 best practices.
Data Cleaning & Preprocessing	10%	Fails to handle nulls, duplicates, invalid codes, or categories.	Basic cleaning done but with little justification or analysis.	All required steps completed correctly with adequate justification.	Thorough, expertly justified, and mindful of biases introduced by cleaning.
Data Analysis & Depth	20%	Superficial or contains significant calculation/interpretation errors.	Basic totals/averages only; lacks depth or business connection.	Correct EDA and risk analysis with valid, logical conclusions.	Deep and insightful; surfaces non-obvious patterns and strong, data-backed hypotheses.
Data Visualization	10%	Missing, misleading, or unreadable plots.	Plots generated but poorly labelled and cluttered.	Clear, correctly labelled, titled, and appropriate plots.	Compelling, polished visuals that tell a story for a non-technical audience.
Report & Communication	30%	Poorly written, unstructured; reads like a code dump.	Summarises notebook output with little business context or strategy.	Well-structured; communicates findings with linked recommendations.	Persuasive, professional narrative with a strong summary and specific, actionable recommendations.
Teamwork & Collaboration	10%	No evidence of collaboration; work clearly siloed.	Minimal collaboration; contributions highly unbalanced.	Clear shared work and meaningful contributions from all members.	Excellent, integrated collaboration with balanced, high-quality contributions.

Part 6: Submission Guidelines

- **Group size:** 4–5 students unless otherwise approved by the lecturer.
- **Submission package:** One compressed folder named **ITS2122_GroupName_FinanceProject.zip**.
- **Contents:** Report PDF, Jupyter Notebook (.ipynb), and the cleaned dataset file.
- **Academic integrity:** All datasets, APIs, external code, and AI-assisted work must be acknowledged. Plagiarism or fabricated outputs may result in zero marks for the affected component.
- **Late submission:** Follows the institute / module late-submission policy announced by the lecturer.
- **Viva readiness:** Each student must be prepared to explain any part of the notebook during the final viva.

References

[1] UCI Machine Learning Repository — Default of Credit Card Clients: https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients

[2] pandas Documentation: https://pandas.pydata.org/docs/

[3] NumPy Documentation: https://numpy.org/doc/

[4] Matplotlib Documentation: https://matplotlib.org/stable/

[5] seaborn Documentation: https://seaborn.pydata.org/

[6] Requests Documentation: https://requests.readthedocs.io/